

Structured Programming Mini Project Ideas

1. SuperShop Grocery Billing System

Description: Manage a supermarket's inventory and customer billing system.

Detailed Functionalities:

- **Inventory Management:**
 - Add new items with a unique ID, name, price, and quantity.
 - Update existing items (e.g., adjust price or quantity).
 - Display all available items in a tabular format.
- **Customer Billing:**
 - Allow customers to select multiple items by ID and specify quantities.
 - Calculate total price, including VAT (e.g., 10%).
- **Stock Updates:**
 - Deduct sold items from inventory automatically.
 - Prevent billing if requested quantity exceeds available stock.
- **Daily Sales Report:**
 - Log details of each transaction (e.g., items sold, quantities, total bill).
 - Display daily sales summary (e.g., total sales, total items sold).

2. Cricket Team Management System

Description: Keep track of a cricket team's players and match performances.

Detailed Functionalities:

- **Player Management:**
 - Add players with details like name, age, role (batsman, bowler, all-rounder), and stats (runs, wickets, matches played).
 - Search players by name or role.
- **Match Stats:**
 - Update performance for a match (runs scored, wickets taken, etc.).
 - Calculate batting average and bowling economy.
- **Team Display:**
 - Show all players in a table, sorted by role or performance.
- **Lineup Generator:**
 - Automatically generate a playing XI based on predefined criteria (e.g., at least 5 batsmen and 4 bowlers).
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3. Bank ATM System

Description: Simulate an ATM system for banking operations.

Detailed Functionalities:

- **Account Management:**
 - Create new accounts with a unique account number, name, PIN, and balance.
 - View account details after verifying the PIN.
- **Transactions:**
 - Deposit money into an account.
 - Withdraw money, ensuring sufficient balance.
 - Display transaction history (e.g., deposits, withdrawals).
- **Secure PIN Handling:**
 - Change PIN securely with proper validation.
- **Error Handling:**
 - Deny transactions for incorrect PIN, invalid account number, or insufficient balance.

4. Library Management System

Description: Manage book inventory and borrowing in a library.

Detailed Functionalities:

- **Book Inventory:**
 - Add new books with title, author, genre, and availability status.
 - Update or remove existing books.
 - Search books by title, author, or genre.
- **Borrow/Return System:**
 - Borrow a book by updating its availability status.
 - Set due dates for borrowed books.
 - Return a book, updating its status and calculating late fees if applicable.
- **Overdue Reports:**
 - Display all overdue books with corresponding borrowers.

5. Student Result Management System

Description: Maintain student records and calculate results.

Detailed Functionalities:

- **Student Records:**
 - Add new student details (ID, name, marks for multiple subjects).
 - Update marks for existing students.
- **Result Calculation:**
 - Compute total marks, average, and grades for each student.
 - Display a ranked list of students by marks.
- **Search and Display:**
 - Search students by ID or name.

- Display individual or all student records.

6. Hospital Patient Management System

Description: Manage patient records in a hospital.

Detailed Functionalities:

- **Patient Records:**
 - Add patient details (name, age, ailment, doctor assigned, status).
 - Update patient status (e.g., admitted, discharged).
- **Search Functionality:**
 - Search patients by name, ailment, or doctor.
- **Daily Report:**
 - Display a list of patients admitted or discharged in a day.
- **Billing System:**
 - Calculate patient bills based on days admitted and treatments received.

7. Event Management System

Description: Organize and manage participants for various events.

Detailed Functionalities:

- **Event Creation:**
 - Add events with details like name, date, location, and description.
 - Update or delete events.
- **Participant Registration:**
 - Register participants with their details (name, contact, and selected event).
 - Search participants by event or name.
- **Event Reports:**
 - Generate reports showing participant counts for each event.

8. Hotel Room Booking System

Description: Manage room bookings in a hotel.

Detailed Functionalities:

- **Room Inventory:**
 - Add rooms with details like room number, type (single, double), and status (available/occupied).
- **Booking System:**
 - Book a room, updating its status and recording customer details.

- Check out a room, calculating charges based on stay duration.
- **Search and Display:**
 - Search for available rooms by type.
 - Display a list of all rooms with current status.

9. Bus Ticket Reservation System

Description: Handle ticket bookings and cancellations for buses.

Detailed Functionalities:

- **Bus Details:**
 - Add buses with details (bus number, route, available seats).
 - Update or remove buses.
- **Ticket Booking:**
 - Reserve tickets by selecting a bus and entering passenger details.
 - Update available seat count after booking.
- **Ticket Cancellation:**
 - Cancel a booking and free up the seat.
- **Reservation Reports:**
 - Display a list of reservations for a specific bus or route.

10. Restaurant Order Management System

Description: Simulate an order and billing system for a restaurant.

Detailed Functionalities:

- **Menu Management:**
 - Add menu items with names, prices, and categories.
 - Update or delete menu items.
- **Order Processing:**
 - Take orders by selecting menu items and quantities.
 - Display the order summary with total cost, including tax and discounts.
- **Daily Sales Report:**
 - Track all orders and generate a summary of total sales.
- **Stock Management:**
 - Deduct ingredients from stock after each order and alert when stock is low.